

our IMU outputs,

$$[\underline{\alpha}]^B = [\underline{a}_B^E]^B - [\underline{g}]^B$$

$$[\underline{\gamma}]^B = [\underline{\omega}_B^E]^B$$

then, the convenient coordination for our dynamics,

$$[D^E \underline{s}_{BT}]^E = [\overline{T}]^{BE} [D^E \underline{s}_{BT}]^B = [\overline{T}]^{BE} [\underline{v}_B^E]^B$$

$$\begin{aligned} [D^E D^E \underline{s}_{BT}]^B &= [D^E \underline{v}_B^E]^B = [\underline{a}_B^E]^B = -[\underline{\Omega}^{BE}]^B [D^E \underline{s}_{BT}]^B + [T]^{BE} [\underline{g}]^E \\ &= -[\underline{\Omega}^{BE}]^B [\underline{v}_B^E]^B + [T]^{BE} [\underline{g}]^E \end{aligned}$$

$$\frac{d}{dt} \begin{pmatrix} \phi \\ \theta \\ \psi \end{pmatrix} = \underline{F}(\phi, \theta) [\underline{\omega}^{BE}]^B$$

$$\frac{d}{dt} \begin{pmatrix} p \\ q \\ r \end{pmatrix} = \left[\frac{d}{dt} \underline{\omega}^{BE} \right]^B = \left([\underline{J}_B^B]^B \right)^{-1} \left([\underline{n}_B]^B - [\underline{\Omega}^{BE}]^B [\underline{J}_B^B]^B [\underline{\omega}^{BE}]^B \right)$$

from 3.20,

$$\left[\frac{d}{dt} T \right]^{EB} = [T]^{EB} [\underline{\Omega}^{BE}]^B$$

where,

$$[\underline{\Omega}^{BE}]^B = [\underline{K}(\underline{\omega}^{BE})]^B = \begin{pmatrix} 0 & -r & q \\ r & 0 & -p \\ -q & p & 0 \end{pmatrix}$$

and $\underline{K}$ is the skew-symmetric operator, and,

$$[\underline{\omega}^{BE}]^B = \begin{pmatrix} p \\ q \\ r \end{pmatrix}$$

and $[T]^{EB}$ is the transpose of the euler transformation matrix $[T]^{BE}$, which is expressed as,

$$[T]^{BE} = \begin{pmatrix} \cos \psi \cos \theta & \sin \psi \cos \theta & -\sin \theta \\ \cos \psi \sin \theta \sin \phi - \sin \psi \cos \varphi & \sin \psi \sin \theta \sin \phi + \cos \psi \cos \phi & \cos \theta \sin \phi \\ \cos \psi \sin \theta \cos \phi + \sin \psi \sin \phi & \sin \psi \sin \theta \cos \phi - \cos \psi \sin \phi & \cos \theta \cos \phi \end{pmatrix}$$

then show,

$$\frac{d}{dt} \begin{pmatrix} \phi \\ \theta \\ \psi \end{pmatrix} = \begin{pmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \frac{\sin \phi}{\cos \theta} & \frac{\cos \phi}{\cos \theta} \end{pmatrix} [\underline{\omega}^{BE}]^B$$